

A Privacy-Preserving Provenance Plane for Reconstructable Fast Healthcare Interoperability Resources to Large Language Model Clinical Artificial Intelligence Transactions

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Abstract—Clinical artificial intelligence systems that consume electronic health record data and emit reviewer-facing summaries rarely preserve the inputs, transformations, model configuration, evidence, safeguards, and human disposition needed to reconstruct a transaction, while storing raw clinical payloads or hidden model reasoning in an immutable log is unsafe. This paper presents Curie Audit Plane, a provenance method for one bounded synthetic Fast Healthcare Interoperability Resources Release 4 to large language model transaction. The plane separates protected clinical content from an append-only audit chain, records typed events with content digests, and verifies hash chaining, Merkle inclusion, and Ed25519 signatures. Structured model output is constrained to rationale, evidence references, uncertainty, assumptions, and missing data. On synthetic fixtures, independently verified audit reconstruction completeness is 20/20 required fields, all 19 labeled tamper cases are detected, and access-control probes pass 11/11 allowed or denied outcomes. Capture overhead is reported against an identical unrecorded clinical workflow, not a thin log shortcut. Significance for biomedical research: the method shows how a FHIR-to-LLM handoff can be reconstructed and independently verified without placing hidden chain-of-thought or raw protected health information in an immutable store.

Index Terms—Biomedical informatics, Fast Healthcare Interoperability Resources, clinical artificial intelligence, provenance, auditability, integrity, health information systems.

I. INTRODUCTION

Large language models are being attached to electronic health record workflows for summarization and review support. Healthcare organizations still need a narrower answer: given one completed transaction, can an investigator reconstruct what the model was shown, how it was configured, which evidence and safeguards were applied, and what a human reviewer decided, without placing raw protected health information or hidden reasoning into an immutable store?

This paper treats that question as a biomedical and health-informatics provenance problem, not as generic software logging. The prototype is deliberately narrow: one synthetic Fast Healthcare Interoperability Resources (FHIR) Release 4 encounter, one structured clinical-summary task, and a human ACCEPT/MODIFY/REJECT disposition.

The frozen research questions are: (i) reconstructability of required provenance fields; (ii) detection of mutation, omis-

sion, reordering, broken references, and invalid proofs; (iii) latency and storage overhead of complete capture versus an identical unrecorded clinical workflow; (iv) console exposure of evidence, uncertainty, guardrail results, and disposition as a system demonstration (SCRIPTED_PROXY), not a blinded reviewer study; and (v) replay fidelity for a deterministic stub versus a hosted nondeterministic model.

II. RELATED WORK

Healthcare provenance and audit have a long FHIR-native literature, including W3C PROV mappings and FHIR Provenance/AuditEvent resources used to record agents, entities, and access [1]–[3]. Those resources are necessary interoperability projections; they are not, by themselves, an internal event chain with independent cryptographic verification.

Clinical decision-support logging and electronic health record audit trails typically record who accessed what, not the full artificial-intelligence transaction: input manifests, transformation records, prompt and model identity, retrieved evidence digests, guardrail outcomes, and the human handoff. Model-provider traces and application JSON logs can retain payloads that should not be immutable, or they omit the links required for reconstruction.

This work is closest to tamper-evident logging and hash-chained audit [4], [5], combined with a clinical-data boundary. It does not replace deterministic alert governance, FHIR access control, or the large language model provider. Related comparisons in the evaluation are application JSON logging, hash-only records without signatures, and FHIR-only projection.

III. METHOD

A. System Boundary

Curie Audit Plane observes a synthetic FHIR-to-large-language-model workflow. Protected payloads (assembled context, model request and response, reviewer-modified text, and free-form comments) are stored in an authorized content store. The immutable audit store holds metadata, opaque references, schemas, and digests. Reviewer comments on the chain are limited to a controlled category; detailed text is protected. Provider exception text is not stored or fingerprinted. Patient and resource identifiers in immutable metadata are opaque tokens; a protected identity map is required for authorized reconstruction.

B. Event Model and Integrity

A successful transaction emits typed events from input manifestation through human action and an integrity-proof commitment. Events are canonically hashed, chained, batched with a Merkle root, and signed. Verification reports VERIFIED, INCOMPLETE, TAMPERED, or FAILED. FHIR Release 4 Provenance and AuditEvent resources are documented projections of the internal model, not a claimed Implementation Guide profile.

C. Structured Output and Guardrails

Model output must validate as structured rationale. Guardrails record schema validity, evidence-reference coverage, uncertainty presence, a synthetic identifier scan, and policy action. Blocking results require an explicit override policy before ACCEPT. Hidden chain-of-thought fields are rejected.

D. Evaluation Protocol

The reproducible command is `curie-audit-plane evaluate`. Reports use schema `curie-evaluation.v1.1` and include experiment metadata (git commit, dirty-tree flag, fixture alias and digest, provider, model, prompt version, decoding parameters, endpoint class, seed, and command template) without absolute local paths as publication identifiers.

Headline audit reconstruction completeness (`independently_verified_arc`) is required-field presence after reloading persisted records, counted only when independent verification succeeds. Field-presence completeness on the in-memory object is reported separately. Tamper detection uses 19 labeled mutation cases. Replay is classified as EXACT_MATCH, EQUIVALENT, or DIVERGENT. Overhead compares the complete plane with an identical unrecorded clinical workflow (load, transform, context, retrieve, complete, guardrails, JSONL) on isolated stores with warmup and three timed repeats:

$$\text{latency_ratio} = (T_{\text{plane}} - T_{\text{no_audit}}) / T_{\text{no_audit}}, \quad (1)$$

$$\text{storage_ratio} = (B_{\text{plane}} - B_{\text{no_audit}}) / B_{\text{no_audit}}. \quad (2)$$

Rate metrics use Wilson intervals; latency ratios use paired differences. Ablations omit input manifests, transformations, model and prompt metadata, evidence references, cryptographic proofs, or human provenance. Access-control probes record allowed and denied HTTP outcomes for reviewer, investigator, admin, output, content, export, missing transaction, and global list scope. Application JSONL and hash-only JSONL are separately instrumented from the unrecorded workflow; the FHIR projection is built from the source bundle. These recorders are not independently shipped products. The complete plane is scored from the signed audit chain.

The default 50-encounter cohort rewrites one synthetic fixture. It is a single-workflow protocol demonstration, not a population sample. Optional Synthea slices may run from approved roots when dumps exist, but they are excluded from measured evidence because generator version, seed, module set, and CLI are not pinned [6].

TABLE I
BASELINE COMPARISON ON THE SYNTHETIC FIXTURE (COMPLETE PLANE VERSUS SEPARATELY INSTRUMENTED UNRECORDED-WORKFLOW AND SOURCE-BUNDLE RECORDERS).

Recorder	ARC	Tamper detection	Queryability
Application JSONL	0.10	0.00	0.00
Hash-only JSONL	0.15	1.00	0.00
FHIR R4 projection	0.20	0.00	0.25
Complete plane	1.00	1.00	1.00

E. Threat Model

Adversaries may mutate, delete, or reorder events; substitute proofs or keys; present unsupported findings; or attempt to place protected health information in reviewer comments, provider errors, research exports, or endpoint configuration. The prototype assumes synthetic data, local keys, and an approved loopback model endpoint. It does not claim protection against compromised signing keys or a malicious protected-content store operator.

IV. RESULTS

All numbers below are from the frozen stub evaluation campaign in `evaluation-results/` (schema `curie-evaluation.v1.1`). Software-quality checks at the time of this draft were 143 Python tests, 94.59% statement coverage, and a clean Ruff pass; those are not clinical performance.

A. Reconstructability and Tamper Detection

Independently verified audit reconstruction completeness is 20/20 required fields on the clean fixture transaction. Field-presence completeness on the in-memory object is also 20/20. Required-event completeness is 11/11 for a successful transaction. All 19 labeled tamper cases are detected, and the false-tamper rate over three clean runs is 0. On the cloned 50-encounter cohort, independently verified completeness remains 1.00 with Wilson 95% interval $[0.93, 1.00]$ ($n = 50$) after reloading persisted records.

The complete plane reconstructs more required fields and detects the standard model-event mutation that application logs and FHIR projections do not detect (Table I). Hash-only logging detects digest mismatch without providing chain or signature guarantees.

B. Ablations

Omitting cryptographic proofs drops reconstructability from 1.00 to 0.75 ($\Delta = 0.25$). Omitting human provenance drops it to 0.85 ($\Delta = 0.15$). Omitting model and prompt metadata drops it to 0.90. Omitting input manifests, transformations, or evidence references each drops one required field (0.95). These are field-presence ablations of recorded provenance classes, not clinical-efficacy ablations (Table II).

TABLE II
RECONSTRUCTABILITY AFTER OMITTING A PROVENANCE CLASS.

Condition	ARC	Δ from full
Full recorded provenance	1.00	0.00
Omit input manifests	0.95	0.05
Omit transformations	0.95	0.05
Omit model metadata	0.90	0.10
Omit evidence references	0.95	0.05
Omit cryptographic proofs	0.75	0.25
Omit human provenance	0.85	0.15

C. Access Control

Eleven role probes passed with observed HTTP status equal to the predeclared expectation (11/11; Wilson 95% interval [0.74, 1.00]): reviewer read and output allowed; reviewer export and investigator output or content denied; investigator verify and export allowed; admin content allowed; unauthenticated list denied (401); missing transaction reported as missing (404); global list allowed.

D. Overhead and Replay

Mean plane latency is 14.78 ms versus 9.15 ms for the identical unrecorded workflow (paired latency ratio 0.615, 95% interval [0.594, 0.637], $n = 3$ after one warmup). Storage ratio versus that workflow is 6.31 (81885 versus 11209 bytes). These values do not support a claim that audit capture is under 15% of an already-complete clinical application. Verification of the clean transaction is 1.07 ms. Deterministic-stub same-prompt replay is exact; prompt-substitution replay is DIVERGENT by construction. Hosted-model same-prompt replay is at best EQUIVALENT and is often DIVERGENT.

The 16-arm scenario matrix exercises AC-CEPT/MODIFY/REJECT, forced and natural WARN/BLOCK, provider failure, sparse encounter, Synthea slices, MODIFY evidence, prompt-substitution replay, and access audit. It is workflow coverage, not a population sample. The two Synthea slice arms are unpinning demonstrations and are excluded from measured tables. Console reconstruction is SCRIPTED_PROXY only.

V. LIMITATIONS

This is a single-workflow protocol demonstration on synthetic FHIR fixtures. The default cohort rewrites one fixture; it is not 50 independent patients and does not support population-style inference. FHIR projections are mapping rules, not Implementation Guide profile validation. Synthea generator version and CLI parameters are NOT_PINNED; optional slice arms are excluded from measured evidence. A DOI for the public archive is not assigned in this draft.

VI. ETHICS, FUNDING, AND DISCLOSURE

No human subjects and no identifiable clinical records were used. The reported evaluation uses synthetic templates and optional external Synthea bundles [6]. Therefore no institutional review board protocol was required. The author reports no specific funding and no conflict of interest. Portions

of software and documentation were drafted with assistance from coding agents; evaluation metrics are produced by the versioned Python harness. The author remains responsible for claims, code, and the decision to submit.

VII. CONCLUSION

A privacy-preserving provenance plane can reconstruct and independently verify a bounded synthetic FHIR-to-large-language-model transaction, detect a defined mutation suite, and keep hidden reasoning and free-form reviewer text out of the immutable chain. Remaining work is experimental scale on independently generated synthetic patients, hosted-model replay characterization, and IEEE Author Portal packaging (PDF Checker, signed consent if required).

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